Towards Efficient Consistency Models via Variance-Reduced Distillation

Useok Choi, Seunjoong Lee, and MyeongAh Cho

***Abstract*—Diffusion models have demonstrated remarkable performance across a wide range of generative tasks; however, their high sampling cost remains a critical bottleneck. To address this, Consistency Distillation was proposed, offering a reduction in sampling cost by distilling a pretrained diffusion model. However, achieving generative quality comparable to diffusion models requires extensive training for distillation process, posing a substantial computational challenge. In this paper, we introduce Variance-Reduced Consistency Learning (vrCL), a novel distillation technique that enables stable and efficient training of consistency models without relying on teacher model evaluations. By leveraging student-guided sample pair, vrCL ensures training stability while significantly reducing computational costs. This design eliminates the need for repeated teacher model evaluations during training, resulting in high computational efficiency and significantly reduced training time. Empirical results demonstrate that vrCL achieves competitive generative performance with high training efficiency, reaching strong results within just 100k training iterations.**

***Index Terms*—Generative models, Diffusion models, Knowledge distillation, Variance reduction, Accelerated training**

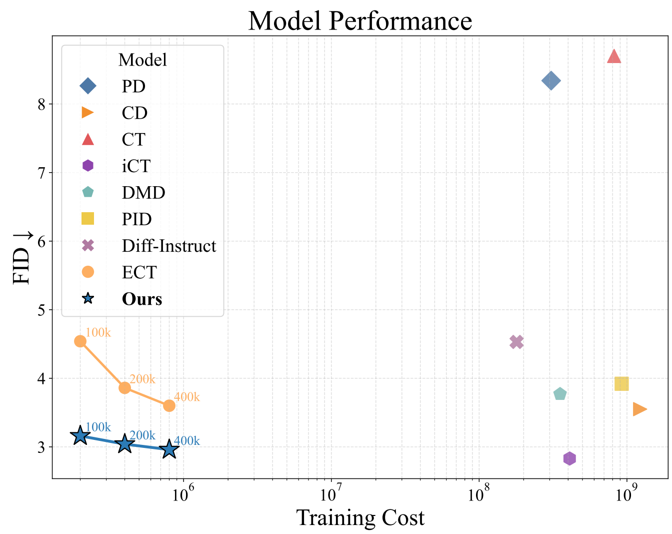

Fig. 1. Comparison of state-of-the-art generative models. Training cost is measured as the product of training images and network evaluation per iteration.

I. Introduction

Diffusion models [1]–[3] have emerged as a powerful class of generative models, leveraging probabilistic flow (PF) Ordinary Differential Equations (ODEs) to model the data distribution through score-based learning [4]. By employing numerical ODE solvers, these models iteratively refine noise into high-quality samples. Despite their impressive performance, diffusion models inherently suffer from high computational costs and error accumulation due to iterative numerical integration. To mitigate these issues, recent advancements in numerical solvers [5]–[9] have reduced the sampling steps to around 10, significantly improving efficiency. These improvements have expanded their practical applications in high-fidelity image synthesis [10]–[15], but the reliance on iterative refinement remains a bottleneck in real-time generative tasks.

To alleviate these issues, recent studies have proposed distillation techniques to enable efficient one-step sampling, eliminating the need for iterative integration. Variance Score Distillation (VSD) techniques [16]–[23] leverage pretrained diffusion models to supervise a one-step generator. These methods typically generate fake samples from a one-step generator G_θ and train a separate fake score network to match the scores of a pretrained diffusion model. Through joint training, the generator G_θ is updated based on feedback from the score matching objective, progressively improving its generative capability. Despite efficiency gains at inference, VSD methods heavily rely on auxiliary networks and often incorporate additional loss terms—such as adversarial losses—which add significant complexity and further dependencies to the training process.

In contrast, Consistency models [24]–[28] have been proposed as an alternative paradigm that enforces self-consistency along the PF ODE trajectory. They train a consistency function that allows direct one-step generation. There are two primary training strategies for Consistency models: Consistency Distillation (CD) and Consistency Training (CT). CD follows a diffusion-supervised approach, utilizes a pretrained diffusion model to approximate the true score function $\nabla \log p(\mathbf{x}_t)$ of data distribution, ensuring training stability by relying on low-variance and accurate score estimation. However, CD requires evaluation from a pretrained diffusion model at every iteration during training, resulting in significant computational overhead and increased training cost.

On the other hand, CT directly optimizes the consistency model using an unbiased score estimator, without relying on approximated scores from a pretrained diffusion model, making it more flexible and computationally efficient. However, CT suffers from high-variance score estimation, leading to unstable training in the early stages, leads to slower convergence compared to CD. These limitations make it challenging for CT to achieve competitive performance without extensive training. Nevertheless, since CT does not rely on a pretrained diffusion model, it avoids potential inaccuracies or biases inherent in the

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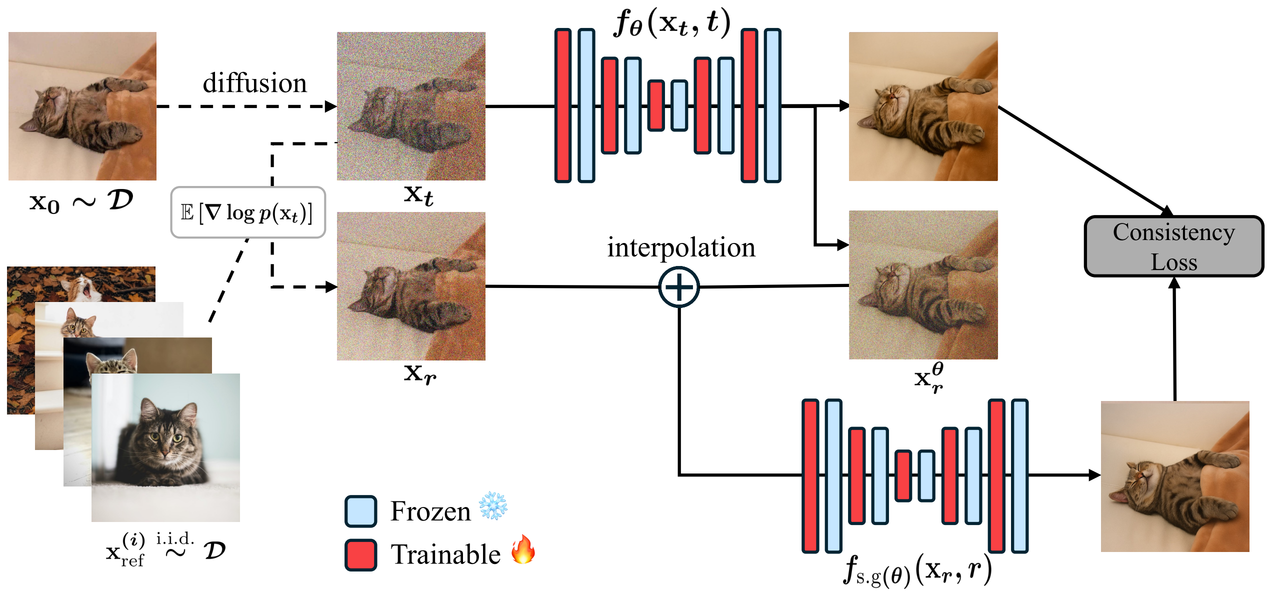


Fig. 2. overview of our Variance-Reduced Consistency Learning (vrCL) framework. We generate a noisy sample $\mathbf{x}_t$ via a forward diffusion process. To construct the $\hat{\mathbf{x}}_r^\theta$, we interpolate between a reference-based stable estimation of $\mathbf{x}_r$, estimated by expected score $\mathbb{E}[\nabla \log p(\mathbf{x}_t)]$, and the student-guided sample $\mathbf{x}_r^\theta$. Reference samples $\mathbf{x}_{\text{ref}}^{(i)}$ are independently drawn from the data distribution $\mathcal{D}$ and are used to estimate expected score. Red and blue blocks denote trainable and frozen network parameters, respectively.

teacher's score estimation. This allows CT to achieve a higher performance upper bound by directly learning consistency function as also observed in recent works [25], [29]–[31].

In this paper, we propose a novel distillation method for training consistency models. Our approach formulates a unified framework that bridges Consistency Distillation (CD) and Consistency Training (CT) through a continuous interpolation mechanism. Rather than a direct combination, it reinterprets the relationship between CD and CT within a single training formulation. Our approach utilizes **two different estimations of $\mathbf{x}_r$** and dynamically interpolates between them during training. The first, $\mathbf{x}_r^\theta$, is a **student-guided reference sample** analogous to that used in CD. Instead of relying on an external teacher model for score supervision like CD, we construct $\mathbf{x}_r^\theta$ directly using the student's own predictions. This eliminates the overhead of repeated forward passes through a pretrained diffusion model by **reusing** student's own predictions $f_\theta(\mathbf{x}_t, t)$ via an implicit ODE traversal, offering greater stability in the early stages of training compared to CT.

As training progresses, we gradually increase reliance on unbiased score estimation $\mathbf{x}_r$. This training strategy inherits the benefits of CT, such as the potential for higher performance than CD, while mitigating the bias that may arise from student-guided reference sample $\mathbf{x}_r^\theta$ in the later stages of training. Through this adaptive strategy, our approach effectively combines the strengths of both consistency training and consistency distillation, enabling efficient and accelerated distillation. Furthermore, we introduce two specialized techniques that enhance score estimation stability and accelerate the distillation process. By integrating these techniques, our framework achieves a computationally efficient training pipeline for high-quality generative modeling.

We empirically show that our method, termed vrCL (Variance-Reduced Consistency Learning), achieves competitive generative performance with significantly reduced training cost requirements on CIFAR-10 and ImageNet 64×64. vrCL achieves strong performance while maintaining low training costs and high learning efficiency. Notably, it attains a 2-step FID of 2.02 on CIFAR-10 within just 100k training iterations.

II. Related Works

Diffusion models [1], [32] have emerged as a dominant paradigm in generative modeling, learning to reverse a forward noising process by estimating the score function of the data distribution [4]. Their success has rapidly expanded across a wide range of domains, including high-fidelity image synthesis [33]–[35], image editing and stylization [11], and even scientific domains such as 3D molecular generation [10]. Despite their high generation quality, they suffer from slow sampling speed, typically requiring tens to thousands of iterative denoising steps.

Various acceleration techniques, such as DDIM [2], higher-order ODE solvers [6], [36], Progressive distillation Technique [17] aim to reduce the sampling steps. However, these methods often exacerbate trajectory mismatch between the training SDE/ODE and the accelerated inference path, resulting in noticeable degradation in perceptual quality.

Consistency models offer an alternative to diffusion models by learning to map noisy samples on the Probabilistic Flow Ordinary Differential Equation (PF-ODE) trajectory directly to their denoised endpoints. The foundational training paradigms include Consistency Distillation (CD), where a pretrained diffusion model provides teacher guidance, and Consistency Training (CT), which learns from scratch without teacher supervision. Early studies CM [24] reported that CT underperforms CD, primarily due to instability when trained without external guidance.

However, recent works [25], [29], [31] have shown that CT can achieve competitive or even superior performance compared to CD. For example, iCT [25] enables student-only training to outperform CD in certain settings, although often at higher computational cost. To mitigate this, ECT [30] reinterprets the denoising score matching (DSM) loss as a generalized form of the consistency objective, leading to a two-phase training framework. While this improves both performance and efficiency, it does not fully resolve the issue of unstable score estimation in CT.

Another line of research focuses on extending consistency models to multi-step generation. CTM [27] introduces a DDIM-inspired parameterization by embedding both the source timestep t and the target timestep s, while PCM [28] segments the PF-ODE trajectory into discrete intervals and performs consistency distillation within each segment. Despite these advancements, such methods often involve more complex training procedures and additional components, which can undermine the simplicity and efficiency of consistency models.

III. Preliminaries

A. Diffusion Models

Diffusion models are a class of probabilistic generative frameworks that model the continuous-time transformation between the data distribution p_{data} and a predefined prior distribution (commonly an isotropic Gaussian). This transformation can be expressed as a stochastic process in forward time, which gradually perturbs the data with Gaussian noise until it becomes indistinguishable from the prior. The reverse of this process corresponds to a denoising generative procedure.

The deterministic counterpart of this stochastic process is described by the *Probability Flow Ordinary Differential Equation* (PF-ODE), which represents the marginal probability evolution without stochasticity:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{f}(\mathbf{x}_t, t) - \frac{1}{2}g^2(t)\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t), \tag{1}$$

where $\mathbf{f}(\mathbf{x}_t, t)$ denotes the drift term determined by the forward process, $g(t)$ is the diffusion coefficient controlling noise injection, and $\nabla \log p(\mathbf{x}_t)$ is the *score function*—the gradient of the log-density of the noisy data distribution at time t.

In practice, a neural network parameterized by θ is trained to approximate this score function. Sample generation is then performed by numerically integrating the PF-ODE backward in time from $t = 1$ (pure noise) to $t = 0$ (clean data). While diffusion models excel in generation quality, they require tens to thousands of integration steps, making them computationally expensive at inference due to their inherently sequential denoising nature.

B. Consistency Models

Consistency models are proposed to overcome the slow sampling bottleneck of diffusion models by bypassing the need for incremental integration. Instead of simulating the PF-ODE step-by-step, a consistency model learns a mapping

$$f_\theta(\mathbf{x}_t, t) \to \mathbf{x}_0$$

that predicts a consistent endpoint (e.g., the clean sample $\mathbf{x}_0$) directly from any noisy state $\mathbf{x}_t$ along the PF-ODE trajectory. This property—known as the *consistency property*—requires that predictions made from different points along the same trajectory agree with each other.

Formally, the consistency matching loss is expressed as:

$$\mathcal{L}^{\text{CM}}(\theta) = \mathbb{E}\big[d\big(f_\theta(\mathbf{x}_t, t), f_{sg(\theta)}(\mathbf{x}_r, r)\big)\big], \tag{2}$$

where $\mathbf{x}_t$ and $\mathbf{x}_r$ ($r < t$) are two states sampled from the same PF-ODE trajectory, $d(\cdot, \cdot)$ denotes a distance metric measuring the discrepancy between two model predictions, and $sg(\theta)$ is the *stop-gradient* operator to prevent trivial solutions. In existing consistency-based frameworks, different forms of $d(\cdot, \cdot)$ have been adopted, including combinations of ℓ_2 and LPIPS losses in early consistency models [24], the pseudo-Huber loss in iCT [25], and adaptive-weighted ℓ_2 objectives in recent approaches such as ECT [30].

There are two primary training paradigms for consistency models, distinguished by whether the student receives guidance from a pretrained diffusion model.

Consistency Distillation (CD): In CD, $\mathbf{x}_t$ is obtained by perturbing a clean sample $\mathbf{x}_0 \sim \mathcal{D}$ with the forward perturbation kernel $q(\mathbf{x}_t|\mathbf{x}_0)$, and $\mathbf{x}_r$ is computed by applying a reverse PF-ODE step using an ODE solver guided by a pretrained diffusion model. This provides low-variance score information and stable target pairs $(\mathbf{x}_t, \mathbf{x}_r)$, which can lead to a more stable training process and faster convergence, but it requires repeated teacher evaluations during each training iteration, resulting in high computational overhead and limited scalability.

Consistency Training (CT): CT Eliminates the teacher model entirely. Here, $\mathbf{x}_r$ is estimated using an unbiased score approximation, e.g.,

$$\mathbf{x}_r \approx \mathbf{x}_0 + r \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

which corresponds to the reverse PF-ODE step guided only by the analytic score estimation. Unlike CD, which computes $\mathbf{x}_r$ by integrating the reverse PF-ODE under a pretrained diffusion teacher at every iterations, CT approximates $\mathbf{x}_r$ without any teacher evaluations. Consequently, CT is computationally efficient and not constrained by the teacher's quality ceiling. However, the unbiased score estimator exhibits high variance, which in turn causes optimization instability and slower convergence relative to CD.

In summary, CD offers stability via external guidance but at the cost of efficiency, whereas CT offers efficiency and potential performance gains but at the cost of stability and requiring more training iterations. This trade-off has motivated numerous recent works to seek hybrid or enhanced training strategies that inherit the strengths of both paradigms.

To better understand the practical implications of using CD versus CT, we provide a detailed quantitative analysis of their training dynamics. While CD benefits from stable supervision, it incurs significant computational cost due to teacher evaluations. In contrast, CT is computationally efficient but may suffer from instability in early training.

For a fair comparison between CD and CT, we initialize both models with a same pretrained diffusion model and apply identical configurations, including timestep scheduling, noise distribution, and other training hyperparameters. We log the 1-step FID score alongside key statistics such as seconds per tick, seconds per kimg, total iterations, and GFLOPS required to reach competitive performance in terms if FID score.

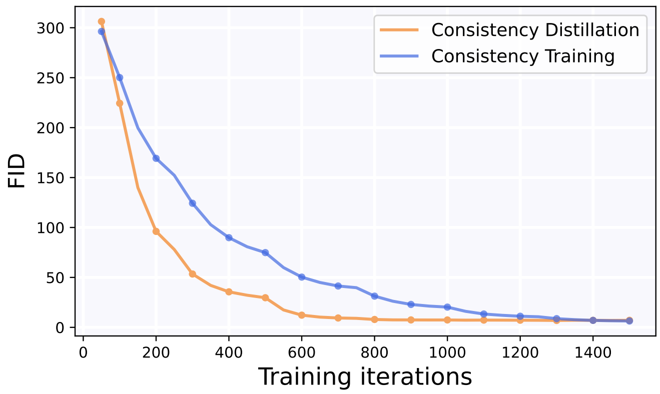


Fig. 3. FID comparison between Consistency Distillation (CD) and Consistency Training (CT) during training on CIFAR-10.

As shown in Fig. 3, CD achieves faster convergence and more stable training compared to CT, reaching a competitive FID within 80k iterations—equivalent in performance to CT at around 135k iterations. However, despite requiring fewer iterations, CD incurs a similar computational cost due to the overhead of evaluating the pretrained diffusion model. This is also reflected in the training time per kimg: CD takes 5.43 seconds, while CT takes only 4.09 seconds.

TABLE I
TRAINING STATISTICS FOR CONSISTENCY DISTILLATION AND CONSISTENCY TRAINING ON CIFAR-10

Method	sec/tick	sec/kimg	Iterations	GFLOPs	FID
CD	66.3	5.43	80,000	5.11M	7.89
CT	52.1	4.09	135,000	5.75M	7.78

Notably, the computational gap between CD and CT becomes more pronounced as model size, dataset scale, or training iteration count increases. Since CD requires repeated evaluations of a large pretrained diffusion model at every step, the overhead scales significantly with training cost. This bottleneck is particularly critical in modern generative modeling pipelines, which employ high-capacity architectures trained on large-scale datasets over hundreds of thousands of iterations. Moreover, recent CD-based consistency models often perform multiple evaluations per step to ensure accurate score estimation, further exacerbating the computational burden. As a result, while CD offers early-stage stability, its high computational demand can limit scalability and practicality in real-world applications. These observations highlight the importance of designing distillation frameworks that retain stability without incurring excessive cost—one of the key motivations behind our proposed method.

IV. VARIANCE-REDUCED CONSISTENCY LEARNING

Existing training methods for consistency models often incur high distillation costs. Consistency Distillation (CD) provides stable supervision via pretrained diffusion models but is computationally expensive. In contrast, Consistency Training (CT) is more efficient but often suffers from instability in the early stages of training. These limitations present a significant bottleneck in the effective training of consistency models. To address this, we propose a simple yet effective training method that combines the efficiency of CT with the stability of CD, thereby bridging the strengths of both approaches.

A. Student-Guided Consistency Distillation

Consistency Distillation (CD) supervises score estimation using reverse-time samples $\mathbf{x}_r$, which are obtained by numerically solving the PF-ODE from $\mathbf{x}_t$ under the guidance of the teacher's score function $s_\phi(\mathbf{x}_t, t) \approx \nabla \log p(\mathbf{x}_t)$. In modern generative modeling pipelines, where high-capacity networks are trained over millions of iterations on large-scale image datasets, the per-step evaluation of a pretrained teacher model poses a major computational burden. To address these limitations and improve the efficiency of consistency model training, we modify the conventional distillation process by removing the reliance on teacher evaluation. Instead, we construct a low-variance reverse-time estimate $\mathbf{x}_r^\theta$ from $\mathbf{x}_t$ using the student model via an Euler- or DDIM [2]-like parameterization of the PF-ODE:

$$\begin{aligned}
\mathbf{x}_r^\theta &= \mathbf{x}_t + \int_t^r \frac{\mathbf{x}_u - \mathbb{E}[\mathbf{x}_0|\mathbf{x}_u]}{u}\, du \\
&\approx \mathbf{x}_t + \int_t^r \frac{\mathbf{x}_t - f_\theta(\mathbf{x}_t, t)}{t}\, du \\
&\approx \mathbf{x}_t + \frac{\mathbf{x}_t - f_\theta(\mathbf{x}_t, t)}{t} \int_t^r du \\
&\approx \frac{r}{t}\mathbf{x}_t + \left(1 - \frac{r}{t}\right) f_\theta(\mathbf{x}_t, t)
\end{aligned} \tag{3}$$

A key advantage of this formulation is that the denoised prediction $f_\theta(\mathbf{x}_t, t)$, used for the consistency matching loss, is also reused to generate $\mathbf{x}_r^\theta$, eliminating the need for costly network evaluations. While CD requires at least three forward passes per step—one for the teacher score evaluation $s_\phi(\mathbf{x}_t, t)$, and two for the student $f_\theta(\mathbf{x}_t, t)$ and $f_{sg(\theta)}(\mathbf{x}_r, r)$—recent generative methods often involve multiple teacher evaluations for score refinement, further increasing the computational cost. In contrast, our method eliminates the need for teacher supervision entirely, requiring only two student evaluations. This matches the computational cost of CT while providing more stable sample pair generation by avoiding high-variance score estimates.

B. Learning with Dual Reference Samples

While our parameterized generation of $\mathbf{x}_r^\theta = \frac{r}{t}\mathbf{x}_t + (1 - \frac{r}{t}) f_\theta(\mathbf{x}_t, t)$ provides a teacher-free mechanism to traverse the PF-ODE trajectory, it may introduce approximation error, as it does not explicitly approximate the score field like diffusion models—which learn the continuous vector field of

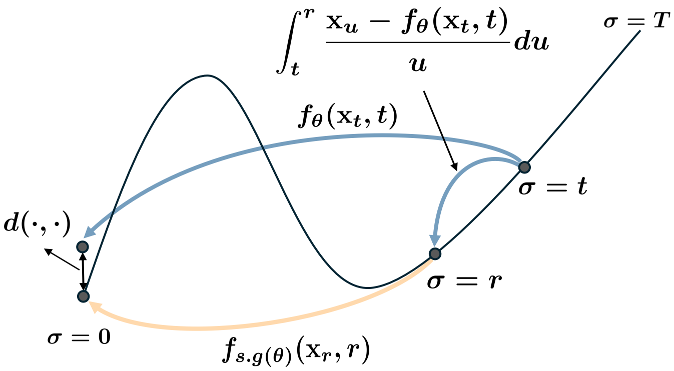


Fig. 4. Illustration of reverse-time reference sample $\mathbf{x}_r^\theta$ generation. Our student-guided parameterization enables $\mathbf{x}_t \to \mathbf{x}_r$ estimation without teacher supervision, offering stable supervision along the PF-ODE trajectory.

Algorithm 1 Variance-Reduced Consistency Learning (vrCL)

Input: Dataset $\mathcal{D}$, pretrained diffusion model ϕ, growth rate β(Iters), weighting function $w(t)$
Initialize: $\theta \leftarrow \phi$, Iters $\leftarrow 0$
repeat
  Sample $\mathbf{x}_0 \sim \mathcal{D},\ \boldsymbol{\epsilon} \sim p(\boldsymbol{\epsilon}),\ t \sim p(t),\ r \sim p(r)$
  Compute $\mathbf{x}_t = \mathbf{x}_0 + t \cdot \boldsymbol{\epsilon},\ \mathbf{x}_r = \mathbf{x}_0 + r \cdot \boldsymbol{\epsilon}$
  Compute $\mathbf{x}_r^\theta = \frac{r}{t}\mathbf{x}_t + (1 - \frac{r}{t}) f_\theta(\mathbf{x}_t, t)$
  Compute $\hat{\mathbf{x}}_r^\theta = \beta \mathbf{x}_r + (1 - \beta)\mathbf{x}_r^\theta$
  Compute loss: $L(\theta) = w(t) \cdot d(f_\theta(\mathbf{x}_t), f_{\text{sg}(\theta)}(\hat{\mathbf{x}}_r^\theta))$
  Update: $\theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$
  Increment: Iters $\leftarrow$ Iters $+ 1$
until convergence
return θ

the PF-ODE—whereas consistency models only learn endpoint mappings. In early training, this formulation remains reliable due to the student being initialized from a pretrained diffusion model. As training progresses, $\mathbf{x}_r^\theta$ may slightly deviate from the ideal trajectory. To mitigate this, we adopt an adaptive strategy that interpolates between two different $\mathbf{x}_r$ estimate: student-guided reference sample $\mathbf{x}_r^\theta$, and unbiased score estimate sample $\mathbf{x}_r$, given by $-(\mathbf{x}_t - \mathbf{x}_0)/t^2$ following CT. While the latter provides a statistically unbiased score supervision signal, it is known to suffer from high variance, which can destabilize training. In contrast, the former offers greater stability, particularly in the early training phase. We define the final training sample $\hat{\mathbf{x}}_r^\theta$ as a convex combination:

$$\hat{\mathbf{x}}_r^\theta = \beta \mathbf{x}_r + (1 - \beta)\mathbf{x}_r^\theta \tag{4}$$

In the early stages of training, $\mathbf{x}_r^\theta$ tends to be more accurate because it leverages the pretrained diffusion model's well-established local denoising capabilities, providing smoother and more stable supervision. As training progresses, however, the unbiased estimate $\mathbf{x}_r$ becomes increasingly precise, capturing the correct global denoising direction without being limited by the student's initial bias. To maintain training stability throughout, we therefore begin with a stronger reliance on $\mathbf{x}_r^\theta$ and gradually shift toward $\mathbf{x}_r$, allowing the model to benefit from both stable early-phase guidance and high-fidelity late-phase refinement.

To maintain training stability throughout, we employ a dynamic scheduling strategy that gradually increases reliance on the unbiased estimate sample $\mathbf{x}_r$ as training progresses. Specifically, we define the effective training reference sample $\hat{\mathbf{x}}_r^\theta$ as a weighted sum between two samples $\mathbf{x}_r, \mathbf{x}_r^\theta$, with the weight adjusted over time by a scheduling function. Inspired by prior works on timestep scheduling in consistency models [25], [30], which have shown that piecewise or staircase-style schedules can stabilize learning dynamics, we adopt a staircase scheduling function. where k is the current training iteration, K is the total number of training iterations, and s is a hyperparameter that controls the growth rate of the schedule. The step interval is defined as $I_K = K/n$, where n denotes the number of growth steps. This scheduling determines how quickly the interpolation shifts from the student-generated reference $\mathbf{x}_r^\theta$ to the unbiased estimate $\mathbf{x}_r$.

$$\beta(k) = \tanh\left(s \cdot \left\lfloor \frac{k}{I_K} \right\rfloor\right), \quad I_K = \frac{K}{n} \tag{5}$$

Although both $\mathbf{x}_r^\theta$ and $\mathbf{x}_r$ approximate the same PF-ODE reverse sample, they offer complementary supervision signals—one derived from model predictions, the other from analytical approximation. This hybrid scheme effectively unifies the benefits of CD and CT: it preserves the stable early-phase dynamics of CD, while ultimately enabling the model to surpass the performance limitations of teacher-driven supervision through the flexible and data-driven generalization of CT. As illustrated in Fig. 7, our method exhibits significantly improved stability in score estimation across training, especially in early stages. We also provide the full algorithmic description of our training procedure in Algorithm 1 to facilitate reproducibility and clarity.

V. Effective Strategies for Variance-Reduced Consistency Learning

To further enhance the compatibility and performance of our proposed distillation framework, we introduce two auxiliary techniques specifically designed to complement our method. The first technique, partial parameter freezing, facilitates rapid distillation by reducing the number of trainable parameters, thereby accelerating convergence and lowering training cost. The second technique addresses the instability that may arise from the use of an unbiased score estimator. We propose a stabilization method that improves the reliability of the unbiased score estimation $-(\mathbf{x}_t - \mathbf{x}_0)/t^2$ during training. Together, these techniques improve both the stability and efficiency of our distillation process.

A. Partial Parameter Freezing for Accelerated Distillation

Freezing layers has been a widely used technique in fine-tuning and transfer learning approaches, as it enables models to retain pretrained knowledge while significantly reducing computational cost during optimization. A recent study [37] has demonstrated the effectiveness of this strategy in the context of score distillation. Pretrained diffusion models already train the convolutional layers to process corrupted images across various noise levels. Therefore, we assume that fully retraining these layers is often unnecessary and can lead to redundant computational cost.

Building on this observation, we apply a selective freezing strategy to our consistency model. Specifically, we freeze the convolutional layers of the backbone—which account for 85.8% of the total parameters—under the assumption that they have already learned rich image representations from prior diffusion model training. Since diffusion models are trained to predict local denoising directions along the trajectory, our consistency model can leverage this pretrained capability and instead focus on learning the global denoising scaling. Thus, only the remaining components are fine-tuned to align frozen features with the consistency objective, reducing the number of trainable parameters, lowering optimization burden, and enabling more efficient and rapid distillation.

This not only speeds up convergence but also mitigates the risk of overfitting when training on limited data or with constrained compute budgets. In practice, we find that this approach achieves comparable or better quality than full fine-tuning, while reducing both memory usage and wall-clock training time.

B. Stabilize Unbiased Score Estimation

In our proposed method, the reverse-time reference sample $\hat{\mathbf{x}}_r^\theta$ is gradually shifted toward the unbiased score estimate sample $\mathbf{x}_r$, which still suffers from high-variance estimation. To address the training variance in later stages of training, we employ a technique that utilizes reference batches for score estimation. Rather than relying on the single-sample estimate $-(\mathbf{x}_t - \mathbf{x}_0)/t^2$, we stabilize score estimation by approximating the conditional expectation $\mathbb{E}\left[\frac{-(\mathbf{x}_t-\mathbf{x}_0)}{t^2}\middle|\mathbf{x}_0\right]$ via Monte Carlo sampling over reference batches.

In contrast to the conventional approach, which estimates the score from a single sample and thus suffers from high variance, our reference-batch strategy leverages multiple samples within each minibatch to estimate the score expectation. By aggregating over diverse $\mathbf{x}_0$ instances, this method produces a more stable and statistically reliable score estimate. As a result, the model benefits from a smoother optimization trajectory and improved convergence stability, particularly in the later training stages when unbiased estimation becomes the dominant guidance signal.

$$\nabla \log p(\mathbf{x}_t) = \mathbb{E}_{p(\mathbf{x}|\mathbf{x}_t)}\left[\nabla \log p(\mathbf{x}_t|\mathbf{x})\right] \tag{6}$$

$$= \mathbb{E}_{p(\mathbf{x})}\left[\frac{p(\mathbf{x}_t|\mathbf{x})}{p(\mathbf{x}_t)}\nabla \log p(\mathbf{x}_t|\mathbf{x})\right] \tag{7}$$

$$\approx \frac{1}{N}\sum_{i=1}^{N}\frac{p(\mathbf{x}_t|\mathbf{x}^{(i)})}{p(\mathbf{x}_t)}\nabla \log p(\mathbf{x}_t|\mathbf{x}^{(i)}) \tag{8}$$

$$= \frac{1}{N}\sum_{i=1}^{N}\frac{p(\mathbf{x}_t|\mathbf{x}^{(i)})}{\frac{1}{N}\sum_{j=1}^{N}p(\mathbf{x}_t|\mathbf{x}^{(j)})}\nabla \log p(\mathbf{x}_t|\mathbf{x}^{(i)}) \tag{9}$$

$$= \frac{1}{N}\sum_{i=1}^{N} w^{(i)}\nabla \log p(\mathbf{x}_t|\mathbf{x}^{(i)}) \tag{10}$$

This technique was originally proposed in the context of diffusion models by [38], and later extended to consistency models by [31], which we follow in our framework. In practice, we implement this by treating each minibatch as a reference batch, allowing us to approximate the expectation without incurring additional computational cost. Specifically, we introduce the normalized importance weight $w^{(i)} = p(\mathbf{x}_t|\mathbf{x}^{(i)})/\frac{1}{N}\sum_{j=1}^{N}p(\mathbf{x}_t|\mathbf{x}^{(j)})$, enabling a weighted aggregation of gradients that emphasizes samples with higher likelihood. When using n reference samples, this Monte Carlo-based approach reduces the estimation variance proportionally to $1/n$, leading to a more reliable training signal.

This strategy is particularly effective in the later stages of training, where the variance of the unbiased score estimation tends to instabilize the training. In the early stage, the model is guided by student-generated samples, while in the later stage, it benefits from the stabilized unbiased estimator—ultimately leading to improved training stability and stronger final performance, as can be observed in the later phase of Fig. 5, where our method maintains a precise score trajectory compared to baselines.

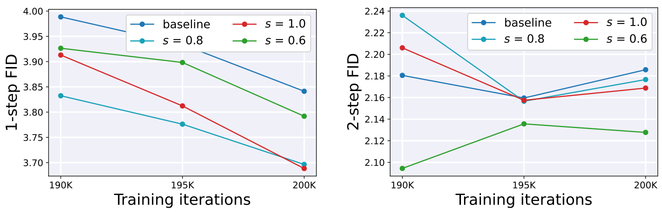
VI. Experimental Results

To evaluate the effectiveness of our proposed method, we conduct a series of comparative experiments using the recent efficient consistency model framework ECT [30] as a baseline. For a fair comparison, we integrate our method on top of baseline without modifying its original configuration. These evaluations aim to highlight the performance and training efficiency improvements enabled by our framework.[1]

A. Implementation Details

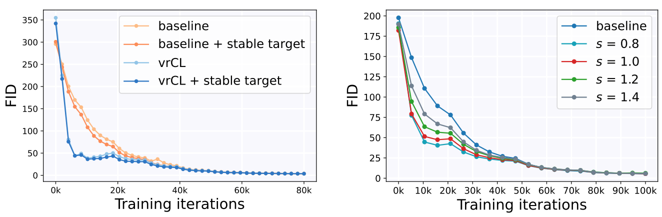
Evaluation Protocol: We evaluate sample quality using Fréchet Inception Distance (FID) [39], computed on 50k samples from both 1-step and 2-step sampling. All experiments follow standard evaluation protocols on CIFAR-10 [40] and ImageNet 64×64 [41], as in [24], [30], [42].

[1]This paper has supplementary downloadable material available at https://github.com/plutoNotebook/variance-reduced-clthe project repository, provided by the authors. This includes the source code, setup scripts, and a README file. This material is 200KB in size.



(a) 1-step sampling results (b) 2-step sampling results

Fig. 5. FID scores (lower is better) under different scheduling configurations for interpolation growth. Our method generally achieves better or competitive performance compared to the baseline across varying hyperparameter s, demonstrating stable performance across the evaluated configurations.



(a) FID Over Iterations (b) FID Curves across s

Fig. 6. Ablation study on training strategies for our model. (Left) Effect of vrCL during training. (Right) Effect of varying s on convergence speed and final FID. For a detailed comparison of final FID results, please refer to Fig. 5

Training Details: For CIFAR-10, we use the DDPM++ [4] architecture with pretrained EDM [42] weights, and for ImageNet 64×64, we use EDM2-(S/M/L/XL) [43] with corresponding pretrained weights. All models are trained with a batch size of 128. Further details and extended results are provided in supplementary material

B. Discussion

Variance Reduced Consistency Learning: To analyze the impact of vrCL in training stability, we compare our method with the baseline. As illustrated in Fig. 6, vrCL achieves a markedly smoother optimization trajectory, reflected in faster and more consistent FID improvements throughout training. This stability is primarily attributed to our variance-reduced score estimation strategy, which mitigates the noisy gradients that often emerge in early training phases.

In addition, Fig. 6 demonstrates that such stability not only accelerates convergence but also consistently translates into superior final performance across a wide range of scheduling intensities, implying that the stable early-phase training provides the model more time to explore and converge toward an optimal solution.

In Fig. 7, we present a direct comparison of score estimation error between vrCL and CT (baseline), measured with respect to the teacher-guided estimation. While the error of CT remains persistently high, vrCL maintains significantly lower error throughout training, with the advantage being most pronounced in the early iterations.

We also investigate the effect of varying scheduling intensity, which controls the rate at which the interpolation

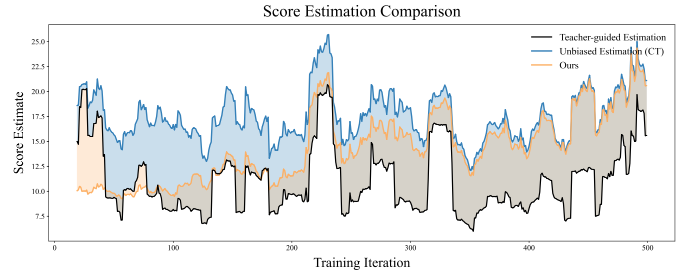


Fig. 7. Comparison of score estimation stability during training. The black curve represents the ground-truth score computed from a pretrained diffusion model, while the blue and yellow curves correspond to the unbiased estimator used in CT and our student-guided estimator, respectively. Shaded areas indicate the absolute estimation error relative to the ground truth. As shown, our method yields lower error than the CT estimator, particularly in early training. In later stages, the error remains suppressed due to the effect of the stabilized estimation technique, demonstrating improved consistency and reliability.

coefficient β shifts from student-guided to unbiased samples. As shown in Fig. 6, higher scheduling intensity (rapid β growth) induces CT-like behavior by quickly prioritizing unbiased supervision, while lower intensity mimics CD-like behavior, retaining more low-variance, student-guided supervision for longer. This tunable balance enables vrCL to adapt between faster convergence and early-phase stability, effectively functioning as a continuous interpolation between the two paradigms—Consistency Distillation and Consistency Training—thereby offering practitioners a principled way to tailor training dynamics to specific datasets and computational budgets.

Stabilize Unbiased Score Estimation: We evaluate the impact of reference-guided unbiased score estimation samples $\mathbf{x}_r$ by incorporating them into both the baseline model and our proposed framework. This technique leverages reference batches to reduce the variance of the unbiased score estimator $\mathbb{E}[-(\mathbf{x}_t - \mathbf{x}_0)/t^2]$ and is implemented using a minibatch as the reference, enabling a more accurate approximation of the expected score without relying on single-sample estimates. As shown in Fig. 7, this approach improves the stability of score estimation in the later stages of training, preventing the performance degradation that often arises from noisy supervision. Moreover, as demonstrated in Fig. 6, it results in more stable overall training dynamics and consistent improvements in FID performance, as further evidenced in Table. V and Table. VI. Importantly, the computation of the expected score using reference samples introduces only negligible additional overhead, as it reuses data already present in the minibatch, making the stability gains effectively cost-free in practical training scenarios.

TABLE II
ABLATION STUDIES OF FREEZING LAYERS ON CIFAR-10.

Norm (7.9%)	Conv (85.8%)	QKV (2.1%)	Skip (4.0%)	FID (↓)
✓	✓	✓	✓	3.63
✓	✗	✓	✓	**3.57**
✓	✗	✗	✓	3.66
✓	✗	✗	✗	3.65

TABLE III
QUANTITATIVE COMPARISON OF SAMPLE QUALITY ON CIFAR-10 DATASET, EVALUATED USING FID.

Unconditional CIFAR-10			
METHOD		**NFE (↓)**	**FID (↓)**
Diffusion models			
Score SDE [4]		2000	2.38
Score SDE (deep) [4]		2000	2.20
EDM [42]		35	**2.04**
Flow Matching [44]		142	6.35
PFGM [45]		110	2.35
Joint Training			
Diffusion GAN [46]		4	3.75
Diffusion StyleGAN [47]		1	3.19
StyleGAN-XL [48]		1	1.52
CTM [27]		1	**1.87**
Diff-Instruct [18]		1	4.53
DMD [22]		1	3.77
SiD [23]		1	1.92
Score Distillation			
DFNO (LPIPS) [49]		1	3.78
2-Rectified Flow [50]		1	4.85
PID (LPIPS) [51]		1	3.92
PD [17]		1	8.34
PD (2-step)		2	5.58
TRACT [52]		1	3.78
TRACT (2-step)		2	3.32
CD (LPIPS) [24]		1	**3.55**
CD (2-step)		2	**2.93**
sCD [29]		1	3.66
sCD (2-step)		2	2.52
Consistency Training	**Iter**	**NFE (↓)**	**FID (↓)**
ACT [53]	-	1	6.4
ACT-Aug [53]	300k	1	6.0
CT [24]	800k	1	8.70
CT (2-step)	800k	2	5.83
iCT [25]	400k	1	2.83
iCT (2-step)	400k	2	2.46
iCT-deep [25]	**400k**	**1**	**2.51**
iCT-deep (2-step)	400k	2	2.24
ECT [30]	100k	1	4.54
ECT (2-step)	100k	2	2.20
vrCL (ours)	100k	1	3.16
vrCL (ours) (2-step)	**100k**	**2**	**2.02**

TABLE IV
QUANTITATIVE COMPARISON OF SAMPLE QUALITY ON IMAGENET 64×64 DATASET, EVALUATED USING FID.

Class-Conditional ImageNet 64×64			
METHOD		**NFE (↓)**	**FID (↓)**
Diffusion models			
ADM [3]		250	2.07
RIN [54]		1000	**1.23**
EDM (Heun) [42]		79	2.44
EDM2 (Heun) [43]		63	1.33
Joint Training			
StyleGAN-XL [48]		1	1.52
Diff-Instruct [18]		1	5.57
EMD [55]		1	2.20
DMD [22]		1	2.62
DMD2 [56]		1	**1.28**
SiD [23]		1	1.52
CTM [27]		1	1.92
Moment Matching [57]		1	3.00
Score Distillation			
DFNO (LPIPS) [49]		1	7.83
PID (LPIPS) [51]		1	9.49
PD [17]		1	15.39
PD (2-step)		2	8.95
TRACT [52]		1	7.43
TRACT (2-step)		2	4.97
CD (LPIPS) [24]		1	6.20
CD (2-step)		2	4.70
MultiStep-CD [58]		1	3.20
MultiStep-CD (2-step)		2	**1.90**
sCD [29]		1	**2.97**
sCD (2-step)		2	2.07
Consistency Training	**Iter**	**NFE (↓)**	**FID (↓)**
ACT [53]	400k	1	10.6
CT (LPIPS) [24]	800k	1	13.0
CT (2-step)	800k	2	11.1
iCT [25]	800k	1	4.02
iCT (2-step)	800k	2	3.20
iCT-deep [25]	**800k**	**1**	**3.25**
iCT-deep (2-step)	**800k**	**2**	**2.77**
ECT [30]	100k	1	5.51
ECT (2-step)	100k	2	3.18
vrCL (ours)	100k	1	5.17
vrCL (ours) (2-step)	100k	2	3.23

Partial Freezing: Motivated by previous findings in score-based generative modeling, we adopt a partial freezing strategy where convolutional layers are fixed during training. We further evaluate the effects of freezing additional components, such as the qkv attention and skip-connection layers, to determine their individual contributions. Table. II shows that, under our proposed vrCL framework, training only a small subset of the model can achieve better performance than full-layer training. This not only simplifies the optimization process but also enables efficient training with significantly fewer iterations. Since the pretrained convolutional layers have already captured rich image features, we retain them as-is and allow the remaining layers to adaptively realign these features to satisfy the consistency property of the model.

C. Comparison with Other Generative Methods

Generative Performance: To obtain optimal performance, we explore the effects of growth rate or β and timestep scheduling. As a result, we observe consistent improvements over the baseline, as demonstrated in Table. III and IV. As

TABLE V
FID (↓) SCORE FOR 1-STEP AND 2-STEP SAMPLING AT DIFFERENT TRAINING ITERATIONS.

Method		100k	200k	400k
1-step	ECT [30]	4.54	3.86	3.60
	Ours	**3.16**	**3.06**	**2.96**
2-step	ECT [30]	2.20	2.15	2.11
	Ours	**2.02**	**2.01**	**2.01**

TABLE VI
COMPARISON OF VRCL WITH BASELINE [30] ON IMAGENET 64×64 ACROSS DIFFERENT EDM2 MODEL SCALES.

Method		EDM2-S	EDM2-M	EDM2-L	EDM2-XL
1-step	ECT [30]	5.51	3.67	3.55	3.35
	Ours	**5.17**	**3.54**	**3.33**	**3.31**
2-step	ECT [30]	**3.18**	2.35	2.14	1.96
	Ours	3.23	**2.18**	**2.11**	1.96

shown Table V, the training time remains consistent with that of baseline, while variance-reduced consistency learning led to significant performance improvements.

Notably, we achieve a 2-step FID of 2.02 on CIFAR-10 within just 100k iterations. In contrast, previous models required significantly more training steps and even larger architectures such as iCT-deep [25]. These results highlight the efficiency of our method in training consistency models while narrowing the gap with joint training approaches. Thanks to the fast convergence and efficient design.

Comparison with the Baseline Across Training Budgets: In addition, to evaluate the efficiency and scalability of our method, we compare its performance with the recent efficient consistency model framework ECT [30], which serves as our baseline, under fixed training budgets of 100k, 200k, and 400k iterations. In these experiments, the **relative speed of the β-shift is kept identical** by fixing the scaling parameters s and n. The results demonstrate that vrCL consistently preserves its performance advantages without degradation even at larger training budgets. Furthermore, as β approaches 1 in the later stage of training, the use of self-referential samples is largely suppressed, indicating that the proposed framework effectively avoids bias accumulation and long-horizon model collapse. Importantly, while the baseline values are directly taken from the ECT paper, we re-tune the timestep scheduling ratio r/t and growth rate β for our method to optimize performance at each budget. As shown in Table V, our method consistently outperforms ECT, with particularly pronounced gains in the low-iteration regime.

In addition, we further validate the scalability of vrCL across model capacities by comparing it with the baseline on multiple EDM2 architectures (EDM2-S/M/L/XL), as summarized in Table VI. Across all model scales and both one-step and two-step sampling settings, vrCL consistently matches or outperforms the baseline, indicating that the proposed variance reduction strategy generalizes well to larger models and does not rely on a specific architectural configuration. These results further demonstrate that the effectiveness of vrCL is preserved not only across training budgets but also across model capacities, suggesting consistent performance under the evaluated settings and indicating its practical applicability.

VII. LIMITATIONS

Our training framework gradually transitions from a pre-trained diffusion model to a consistency model via student-guided reverse-time references. Prolonged reliance on these student-generated references in later stages may introduce bias, as the model increasingly reflects global dynamics rather than local PF-ODE behavior. To alleviate this issue, we carefully control the growth rate of β. This added complexity is modest compared to the overall hyperparameter scale of modern generative models. In addition, we note that our experimental validation is primarily conducted at lower resolutions (e.g., 32×32 and 64×64), and evaluation at higher resolutions is not included in the current work. While we expect the proposed framework to generalize to higher resolutions, verifying this remains an important direction for future work.

VIII. CONCLUSION

We propose Variance-Reduced Consistency Learning (vrCL), a unified training framework for consistency models that eliminates the need for evaluating pretrained diffusion models during training while maintaining stability and efficiency. By integrating the strengths of Consistency Distillation and Consistency Training, vrCL constructs reverse-time samples via student-guided parameterization and interpolates them with unbiased estimates, enabling low-variance supervision without additional architectural complexity. Our framework preserves the simplicity of one-step decoding, supports scalability across diverse generative settings, and reduces training cost through a selective freezing strategy. In addition, minibatch-based score estimation further stabilizes the supervision signal, making vrCL an efficient and practical approach for stable, teacher-free generative modeling.

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REFERENCES

[1] B. B. Moser, A. S. Shanbhag, F. Raue, S. Frolov, S. Palacio, and A. Dengel, "Diffusion models, image super-resolution, and everything: A survey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 7, pp. 11 793–11 813, 2025.

[2] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *International Conference on Learning Representations (ICLR)*, 2021.

[3] P. Dhariwal and A. Nichol, "Diffusion models beat gans on image synthesis," *Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.

[4] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," *arXiv preprint arXiv:2011.13456*, 2020.

[5] Z. Kong and W. Ping, "On fast sampling of diffusion probabilistic models," *arXiv preprint arXiv:2106.00132*, 2021.

[6] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, "Dpm-solver: A fast ode solver for diffusion probabilistic model sampling in around 10 steps," *arXiv preprint arXiv:2206.00927*, 2022.

[7] Q. Zhang and Y. Chen, "Fast sampling of diffusion models with exponential integrator," *arXiv preprint arXiv:2204.13902*, 2022.

[8] Z. Zhou, D. Chen, C. Wang, and C. Chen, "Fast ode-based sampling for diffusion models in around 5 steps," *arXiv preprint arXiv:2312.00094*, 2023.

[9] S. Xue, M. Yi, W. Luo, S. Zhang, J. Sun, Z. Li, and Z.-M. Ma, "Sa-solver: Stochastic adams solver for fast sampling of diffusion models," *Neural Information Processing Systems (NeurIPS)*, 2024.

[10] H. Huang, L. Sun, B. Du, and W. Lv, "Learning joint 2-d and 3-d graph diffusion models for complete molecule generation," *IEEE transactions on neural networks and learning systems*, vol. 35, no. 9, pp. 11 857–11 871, 2024.

[11] N. Huang, Y. Zhang, F. Tang, C. Ma, H. Huang, W. Dong, and C. Xu, "Diffstyler: Controllable dual diffusion for text-driven image stylization," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, no. 2, pp. 3370–3383, 2024.

[12] Z. Jiang, G. Fang, J. Han, G. Lu, H. Xu, S. Liao, X. Chang, and X. Liang, "Realigndiff: Boosting text-to-image diffusion model with coarse-to-fine semantic realignment," *IEEE Transactions on Neural Networks and Learning Systems*, 2025.

[13] C. Saharia, J. Ho, W. Chan, T. Salimans, D. J. Fleet, and M. Norouzi, "Image super-resolution via iterative refinement," *IEEE transactions on pattern analysis and machine intelligence*, vol. 45, no. 4, pp. 4713–4726, 2022.

[14] A. Hatamizadeh, J. Song, G. Liu, J. Kautz, and A. Vahdat, "Diffit: Diffusion vision transformers for image generation," *arXiv preprint arXiv:2312.02139*, 2023.

[15] A. Ramesh, M. Pavlov, G. Goh, S. Gray, C. Voss, A. Radford, M. Chen, and I. Sutskever, "Zero-shot text-to-image generation," in *International Conference on Machine Learning (ICML)*, 2021.

[16] E. Luhman and T. Luhman, "Knowledge distillation in iterative generative models for improved sampling speed," *ArXiv*, vol. abs/2101.02388, 2021.

[17] T. Salimans and J. Ho, "Progressive distillation for fast sampling of diffusion models," *arXiv preprint arXiv:2202.00512*, 2022.

[18] W. Luo, T. Hu, S. Zhang, J. Sun, Z. Li, and Z. Zhang, "Diff-instruct: A universal approach for transferring knowledge from pre-trained diffusion models," *Neural Information Processing Systems (NeurIPS)*, 2024.

[19] J. Gu, S. Zhai, Y. Zhang, L. Liu, and J. M. Susskind, "Boot: Data-free distillation of denoising diffusion models with bootstrapping," in *ICML 2023 Workshop on Structured Probabilistic Inference* {\&} *Generative Modeling*, 2023.

[20] A. Sauer, D. Lorenz, A. Blattmann, and R. Rombach, "Adversarial diffusion distillation," *arXiv preprint arXiv:2311.17042*, 2023.

[21] Z. Geng, A. Pokle, and J. Z. Kolter, "One-step diffusion distillation via deep equilibrium models," *Neural Information Processing Systems (NeurIPS)*, vol. 36, 2024.

[22] T. Yin, M. Gharbi, R. Zhang, E. Shechtman, F. Durand, W. T. Freeman, and T. Park, "One-step diffusion with distribution matching distillation," *arXiv preprint arXiv:2311.18828*, 2023.

[23] M. Zhou, H. Zheng, Z. Wang, M. Yin, and H. Huang, "Score identity distillation: Exponentially fast distillation of pretrained diffusion models for one-step generation," in *International Conference on Machine Learning (ICML)*, 2024.

[24] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever, "Consistency models," *arXiv preprint arXiv:2303.01469*, 2023.

[25] Y. Song and P. Dhariwal, "Improved techniques for training consistency models," *arXiv preprint arXiv:2310.14189*, 2023.

[26] S. Luo, Y. Tan, S. Patil, D. Gu, P. von Platen, A. Passos, L. Huang, J. Li, and H. Zhao, "Lcm-lora: A universal stable-diffusion acceleration module," *arXiv preprint arXiv:2311.05556*, 2023.

[27] D. Kim, C.-H. Lai, W.-H. Liao, N. Murata, Y. Takida, T. Uesaka, Y. He, Y. Mitsufuji, and S. Ermon, "Consistency trajectory models: Learning probability flow ODE trajectory of diffusion," in *International Conference on Learning Representations (ICLR)*, 2024.

[28] F.-Y. Wang, Z. Huang, A. W. Bergman, D. Shen, P. Gao, M. Lingelbach, K. Sun, W. Bian, G. Song, Y. Liu *et al.*, "Phased consistency model," *arXiv preprint arXiv:2405.18407*, 2024.

[29] C. Lu and Y. Song, "Simplifying, stabilizing and scaling continuous-time consistency models," *arXiv preprint arXiv:2410.11081*, 2024.

[30] Z. Geng, A. Pokle, W. Luo, J. Lin, and J. Z. Kolter, "Consistency models made easy," *arXiv preprint arXiv:2406.14548*, 2024.

[31] F.-Y. Wang, Z. Geng, and H. Li, "Stable consistency tuning: Understanding and improving consistency models," *arXiv preprint arXiv:2410.18958*, 2024.

[32] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, and S. Ganguli, "Deep unsupervised learning using nonequilibrium thermodynamics," in *International Conference on Machine Learning (ICML)*, 2015.

[33] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models. 2022 ieee," in *CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 10 674–10 685.

[34] C. Saharia, W. Chan, S. Saxena, L. Li, J. Whang, E. L. Denton, K. Ghasemipour, R. Gontijo Lopes, B. Karagol Ayan, T. Salimans *et al.*, "Photorealistic text-to-image diffusion models with deep language understanding," *Advances in neural information processing systems*, vol. 35, pp. 36 479–36 494, 2022.

[35] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, "Hierarchical text-conditional image generation with clip latents," *arXiv preprint arXiv:2204.06125*, vol. 1, no. 2, p. 3, 2022.

[36] C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, and J. Zhu, "Dpm-solver++: Fast solver for guided sampling of diffusion probabilistic models," *arXiv preprint arXiv:2211.01095*, 2022.

[37] B. Zheng and T. Yang, "Diffusion models are innate one-step generators," *arXiv preprint arXiv:2405.20750*, 2024.

[38] Y. Xu, S. Tong, and T. S. Jaakkola, "Stable target field for reduced variance score estimation in diffusion models," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=WmIwYTd0YTF

[39] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "Gans trained by a two time-scale update rule converge to a local nash equilibrium," *Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[40] A. Krizhevsky, "Learning multiple layers of features from tiny images," 2009. [Online]. Available: https://www.cs.toronto.edu/~kriz/cifar.html

[41] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE conference on computer vision and pattern recognition*. Ieee, 2009, pp. 248–255.

[42] T. Karras, M. Aittala, T. Aila, and S. Laine, "Elucidating the design space of diffusion-based generative models," in *Proc. NeurIPS*, 2022.

[43] T. Karras, M. Aittala, J. Lehtinen, J. Hellsten, T. Aila, and S. Laine, "Analyzing and improving the training dynamics of diffusion models," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[44] Y. Lipman, R. T. Chen, H. Ben-Hamu, M. Nickel, and M. Le, "Flow matching for generative modeling," *arXiv preprint arXiv:2210.02747*, 2022.

[45] Y. Xu, Z. Liu, M. Tegmark, and T. Jaakkola, "Poisson flow generative models," *Advances in Neural Information Processing Systems*, vol. 35, pp. 16 782–16 795, 2022.

[46] Z. Xiao, K. Kreis, and A. Vahdat, "Tackling the generative learning trilemma with denoising diffusion GANs," in *International Conference on Learning Representations (ICLR)*, 2022. [Online]. Available: https://openreview.net/forum?id=JprM0p-q0Co

[47] Z. Wang, H. Zheng, P. He, W. Chen, and M. Zhou, "Diffusion-gan: Training gans with diffusion," *arXiv preprint arXiv:2206.02262*, 2022.

[48] A. Sauer, K. Schwarz, and A. Geiger, "Stylegan-xl: Scaling stylegan to large diverse datasets," in *ACM SIGGRAPH 2022 conference proceedings*, 2022.

[49] H. Zheng, W. Nie, A. Vahdat, K. Azizzadenesheli, and A. Anandkumar, "Fast sampling of diffusion models via operator learning," in *International Conference on Machine Learning (ICML)*. PMLR, 2023.

[50] X. Liu, C. Gong, and Q. Liu, "Flow straight and fast: Learning to generate and transfer data with rectified flow," *arXiv preprint arXiv:2209.03003*, 2022.

[51] J. T. J. Tee, K. Zhang, H. S. Yoon, D. N. Gowda, C. Kim, and C. D. Yoo, "Physics informed distillation for diffusion models," *Transactions on Machine Learning Research*, 2024. [Online]. Available: https://openreview.net/forum?id=rOvaUsF996

[52] D. Berthelot, A. Autef, J. Lin, D. A. Yap, S. Zhai, S. Hu, D. Zheng, W. Talbott, and E. Gu, "Tract: Denoising diffusion models with transitive closure time-distillation," *arXiv preprint arXiv:2303.04248*, 2023.

[53] F. Kong, J. Duan, L. Sun, H. Cheng, R. Xu, H. Shen, X. Zhu, X. Shi, and K. Xu, "Act-diffusion: Efficient adversarial consistency training for one-step diffusion models," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 8890–8899.

[54] A. Jabri, D. Fleet, and T. Chen, "Scalable adaptive computation for iterative generation," *arXiv preprint arXiv:2212.11972*, 2022.

[55] S. Xie, Z. Xiao, D. P. Kingma, T. Hou, Y. N. Wu, K. P. Murphy, T. Salimans, B. Poole, and R. Gao, "Em distillation for one-step diffusion models," *arXiv preprint arXiv:2405.16852*, 2024.

[56] T. Yin, M. Gharbi, T. Park, R. Zhang, E. Shechtman, F. Durand, and W. T. Freeman, "Improved distribution matching distillation for fast image synthesis," *arXiv preprint arXiv:2405.14867*, 2024.

[57] T. Salimans, T. Mensink, J. Heek, and E. Hoogeboom, "Multistep distillation of diffusion models via moment matching," *arXiv preprint arXiv:2406.04103*, 2024.

[58] J. Heek, E. Hoogeboom, and T. Salimans, "Multistep consistency models," *arXiv preprint arXiv:2403.06807*, 2024.

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